Core Pure 1 · Chapter 8

Proof by Induction

Summation · Divisibility · Matrix powers

What is Proof by Induction?

Proof by induction establishes a statement for every integer from a starting value onwards.

Example: prove

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$$

for all positive integers n.

Checking $n = 3$:

$$1 + 2 + 3 = 6 = \frac{1}{2}(3)(4)$$

This confirms one case only. No finite list of examples proves the statement for all positive integers. We need a general argument.

How it works

Luisa · 3

Given $a_1 = 3$ and $a_{n+1} = a_n + 2$, prove that every term is odd.

Basis

Inductive step

Conclusion

Think of a line of dominoes: push the first, and show that each falling domino knocks over the next.

Chapter Overview

Luisa · 4

1. **Summation proofs**

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$$

2. **Divisibility proofs**

Prove $n^3 - 7n + 9$ is divisible by 3.

3. **Matrix proofs**

$$\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}^n = \begin{pmatrix} 1 & 1-2^n \\ 0 & 2^n \end{pmatrix}$$

Throughout this chapter, $n \in \mathbb{Z}^+$: the positive integers.

1. **Basis**

Prove the statement for the starting value, usually $n = 1$.

2. **Assumption**

Assume it holds for an arbitrary positive integer $n = k$.

3. **Inductive step**

Use that assumption to prove the statement for $n = k + 1$.

4. **Conclusion**

State why the basis and inductive step together prove all positive integer cases.

Prove

$$\sum_{r=1}^{n}(2r-1) = n^2$$

1. Basis

3. Induction

2. Assumption

4. Conclusion

A complete conclusion

The result holds for $n = 1$, and truth for $n = k$ implies truth for $n = k + 1$.
Therefore it holds for every positive integer n, by induction.

Make the logical chain explicit:

1. The statement is true at the starting value.
2. You assumed it for an arbitrary k.
3. You proved the next case from that assumption.
4. It therefore follows for all positive integers.

Equivalent clear wording is acceptable.

Prove

$$\sum_{r=1}^{n} r2^r = 2[1 + (n-1)2^n]$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Test Your Understanding

Prove by induction

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Exercise 8A - Core Pure 1 Pages 158-159

1. Prove

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Prove by induction

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

3a. Prove

$$\sum_{r=1}^{n} r(r-1) = \frac{1}{3}n(n+1)(n-1)$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

3b. Hence find $\sum_{r=1}^{2n+1} r(r-1)$.

b

4a. Prove

$$\sum_{r=1}^{n} r(3r-1) = n^2(n+1)$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

4b. Find positive n if $\sum r^3 = 4\sum r(3r-1)$, both sums from 1 to n.

b

5a. Prove

$$\sum_{r=1}^{n} (\frac{1}{2})^r = 1 - 2^{-n}$$

.

5b. Prove

$$\sum_{r=1}^{n} r(r!) = (n+1)! - 1$$

.

5c. Let

$$S_n = \sum_{r=1}^{n} \frac{4}{r(r+2)}.$$

Prove

$$S_n = \frac{n(3n+5)}{(n+1)(n+2)}.$$

a

b

c

Conclusion

6. A student claims

$$\left(\sum_{r=1}^{n} r\right)^2 = \sum_{r=1}^{n} r^2$$

They check $n = 1$, assume the statement at k, then simply write the statement at $k + 1$ and conclude it is true.

a. Identify the error.

b. Give a counterexample.

a

b

7. A student claims

$$\sum_{r=1}^{n} r = \frac{1}{2}(n^2 + n + 1)$$

They assume the formula at k, add $k + 1$ to both sides and correctly obtain the formula at $k + 1$, then conclude it is true.

a. Identify the error.

b. Give a counterexample.

a

b

Challenge: prove

$$\sum_{r=1}^{n}(-1)^r r^2 = \frac{1}{2}(-1)^n n(n+1)$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Type 2: Divisibility Proofs **Luisa · 18**

Prove $f(n) = 3^{2n} + 11$ is divisible by 4 for every positive integer n.

1. Basis

2. Assumption

Continue with the inductive step on the next slide.

Method 1: compare successive terms

Let $f(n) = 3^{2n} + 11$. The basis and assumption were established on the preceding slide.

3. Induction

4. Conclusion

Method 2: split off the assumed expression

With the same basis and assumption, rewrite $f(k+1)$ directly.

Induction and conclusion

Type 2: Divisibility Proofs

Luisa · 21

Prove $f(n) = 8^n - 3^n$ is divisible by 5 for every positive integer n.

1. Basis

2. Assumption

3. Induction

4. Conclusion

Test Your Understanding

Luisa · 22

Prove $f(n) = n^3 - 7n + 9$ is divisible by 3 for every positive integer n.

1. Basis

2. Assumption

3. Induction

4. Conclusion

Let $f(n) = 13^n - 6^n$.

a. Show that $f(k+1) = 6f(k) + 7 \cdot 13^k$.

b. Prove by induction that $f(n)$ is divisible by 7 for every positive integer n.

1. Basis

2. Assumption

3. Induction

4. Conclusion

Let $f(n) = 5^{2n} - 6n + 8$.

a. Show that $f(k+1) = 25f(k) + 9(16k - 22)$.

b. Prove by induction that $f(n)$ is divisible by 9 for every positive integer n.

1. Basis

2. Assumption

3. Induction

4. Conclusion

Exercise 8B Core Pure 1 Page 162 — Luisa · 25

Prove $f(n) = 8^n - 3^n$ is divisible by 5 for every positive integer n.

1. Basis

2. Assumption

3. Induction

4. Conclusion

Exercise 8B Core Pure 1 Page 162 — Luisa · 26

Prove $f(n) = 3^{2n+2} + 8n - 9$ is divisible by 8 for every positive integer n.

1. Basis

2. Assumption

3. Induction

4. Conclusion

Exercise 8B Core Pure 1 Page 162 — Luisa · 27

Prove $f(n) = 2^{6n} + 3^{2n-2}$ is divisible by 5 for every positive integer n.

1. Basis

2. Assumption

3. Induction

4. Conclusion

Type 3: Matrix Proofs — Luisa · 28

Let

$$A = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 1 & 1-2^n \\ 0 & 2^n \end{pmatrix}$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Test Your Understanding

Let

$$A = \begin{pmatrix} -2 & 9 \\ -1 & 4 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 1-3n & 9n \\ -n & 3n+1 \end{pmatrix}$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Exercise 8C - Core Pure 1 Page 164

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Exercise 8C - Core Pure 1 Page 164 — Luisa · 31

Let

$$A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 2n+1 & -4n \\ n & 1-2n \end{pmatrix}$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Exercise 8C - Core Pure 1 Page 164 — Luisa · 32

Let

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 2^n & 0 \\ 2^n - 1 & 1 \end{pmatrix}$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

Let

$$A = \begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 4n+1 & -8n \\ 2n & 1-4n \end{pmatrix}$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

b. Find n if

$$\begin{pmatrix} -1 & 3 \\ -2 & 5 \end{pmatrix} A^n = \begin{pmatrix} 11 & -21 \\ 10 & -19 \end{pmatrix}$$

b

Let

$$A = \begin{pmatrix} 2 & 5 \\ 0 & 1 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 2^n & 5(2^n-1) \\ 0 & 1 \end{pmatrix}$$

1. Basis

2. Assumption

3. Induction

4. Conclusion

b. Find $(A^n)^{-1}$.

b

Let

$$A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 4 \end{pmatrix}$$

Prove for every positive integer n:

$$A^n = \begin{pmatrix} 3^n & \frac{3^n-1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & \frac{1-4^n}{3} & 4^n \end{pmatrix}$$

1. Basis

2. Assumption

3. Induction

4. Conclusion